Using the physical-isomorphisms crossovers - five workload decisions computed from a TCT output

TCT applied study

Using the physical-isomorphisms crossovers — five workload decisions computed from a TCT output

Companion artifact to `case_studies/physical-isomorphisms-tct` (study 2.3, "how can physical functions be mapped as isomorphisms to computational systems, with measurable improvement") · **Date** 2026-11-01 · **Method** every verdict below is computed exactly, in `Fraction`/`int`, from the case study's re-verified premises — the TCT discipline carried into application (`scripts/tct_applied_decisions.py`, log in `tct_applied/decisions.log`).

1. The test

The case study ends, like every TCT run, with measured constants and honest caveats. This brief is the obvious next question: **can those outputs be used?** Not quoted — used, the way an engineer uses a datasheet: enter the workload parameters, get a verdict with its exact crossover location and its caveat attached. The test below takes five concrete workload decisions, feeds each through the case study's cost models, and reports what comes out. The deliverables are this brief, the decision log, and a reusable instrument — a self-contained HTML substrate selector whose verdict logic is parity-tested against the Python engine (§9).

One framing note. The case study's constants are of two kinds, and the verdicts inherit the distinction: **empirical premises** (the Horowitz 2014 energy table: 37-unit multiply, 50-unit SRAM read, 6400-unit DRAM read) and **declared illustrative model values** (crossbar cell = 2 units, converter = 3700 units, analog floor = 1/1000). The Lean tier of the study records exactly this split — six axioms, three sorry lemmas. Decisions computed from axioms are decisions about the model; they are as strong as the model, and no stronger. That inheritance is itself part of what "using a TCT output" means: the epistemic hygiene ships with the numbers.

2. The premises used

All six Column-3 scripts of the case study were re-run live on 2026-11-01 before any decision was computed (`tct_applied/reverification.log`); every headline constant reproduced exactly.

Premise	Value	Source	Status
Per-MAC energy, von Neumann	12838 units (0.1 pJ)	script 01	empirical (Horowitz 2014)
Per-MAC energy, in-memory	138 units	script 01	empirical (Horowitz 2014)
Crossbar cell / converter	2 / 3700 units	script 01	illustrative (axiom)
Optical 4f: setup / per-frame	1048576 / 1025 ops	script 02	illustrative (axiom)
FFT per frame (N=1024)	51200 ops	script 02	derived, exact
Chebyshev stream length	$N \geq p(1-p)/\varepsilon^2$	script 03	theorem (carried)
AND gate vs FP multiply	1 vs 37 units	script 03	empirical
Analog error floor	1/1000	script 04	illustrative (axiom)
Euler error	$e_M = (3M-1)/2M^2$	script 04	derived, exact
SAT→Ising mapping	exact, 8/8 assignments	script 06	derived, exact
Extended Church–Turing	—	script 06 / ledger	axiom (`physical_church_turing`)

The Lean ledger of the study scans, as before, **3 sorry** (`break_even_symbolic`, `euler_error_general`, `stochastic_n_required`) and **6 axioms** — the development the study still owes and the premises it honestly assumes, unchanged by this brief.

3. Decision A — an edge-inference matmul engine (N = 1024)

Question. A 1024×1024 matrix–vector engine for edge inference: does an analog crossbar beat digital, and by how much, at 8-bit and at 32-bit effective precision?

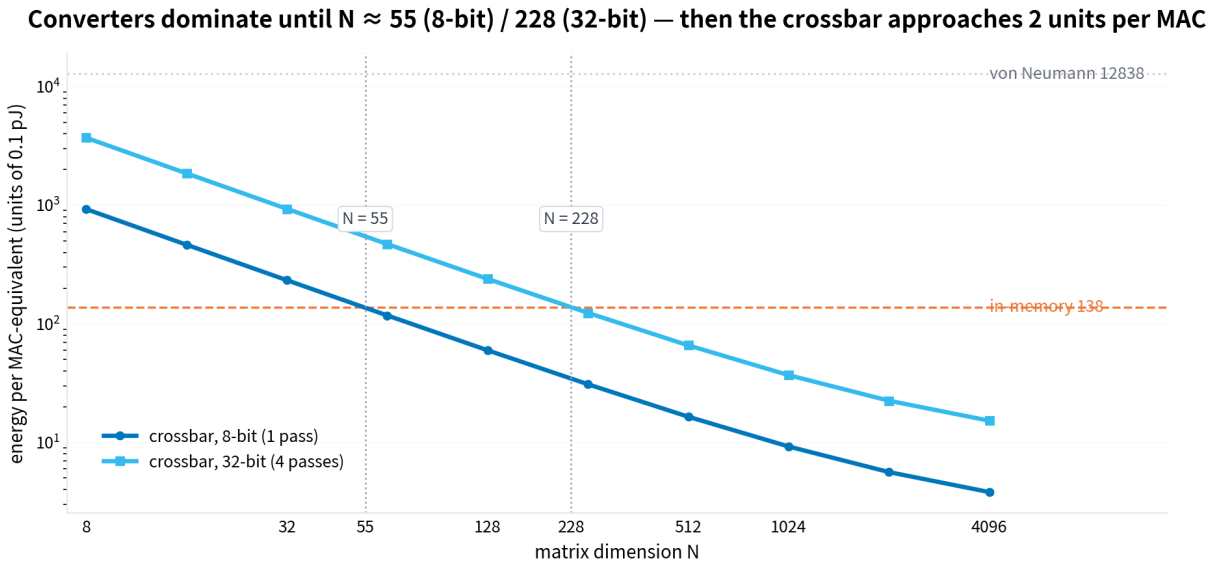
Computation. Analog total = $P\cdot(2N^2 + 7400N)$ with P = bits/8 passes; digital totals are $138\cdot N^2$ (in-memory) and $12838\cdot N^2$ (von Neumann). At N = 1024:

Precision	Analog total	vs in-memory	vs von Neumann
8-bit (1 pass)	9,674,752 units	17664/1181 ≈ **14.96×** fewer	1643264/1181 ≈ **1391.42×** fewer
32-bit (4 passes)	38,699,008 units	4416/1181 ≈ **3.74×** fewer	410816/1181 ≈ **347.85×** fewer

Verdict. The crossbar wins at N = 1024 under every comparison, but the margin collapses with precision: bit-slicing to 32 bits (four passes) eats four× the converter overhead and shrinks a 15× win to 3.7× against in-memory digital. The asymptotic 69× is the converter-free limit and is never reached at practical N.

Location and caveat. The break-even against in-memory digital sits at $N > 2960/13$, i.e. $N \geq 228$ for 32-bit and $N \geq 55$ for 8-bit — N = 1024 clears both (Fig. 1). Below

those sizes the converters eat the compute: this is precondition P4 (I/O discipline) of the study, now with a number attached. Constants illustrative; the verdict is about the model.



Per-MAC energy versus matrix size N. The crossbar's effective per-MAC cost falls toward the 2-unit cell as converters amortize; digital baselines are flat. Break-evens at N = 55 (8-bit) and N = 228 (32-bit) against in-memory.

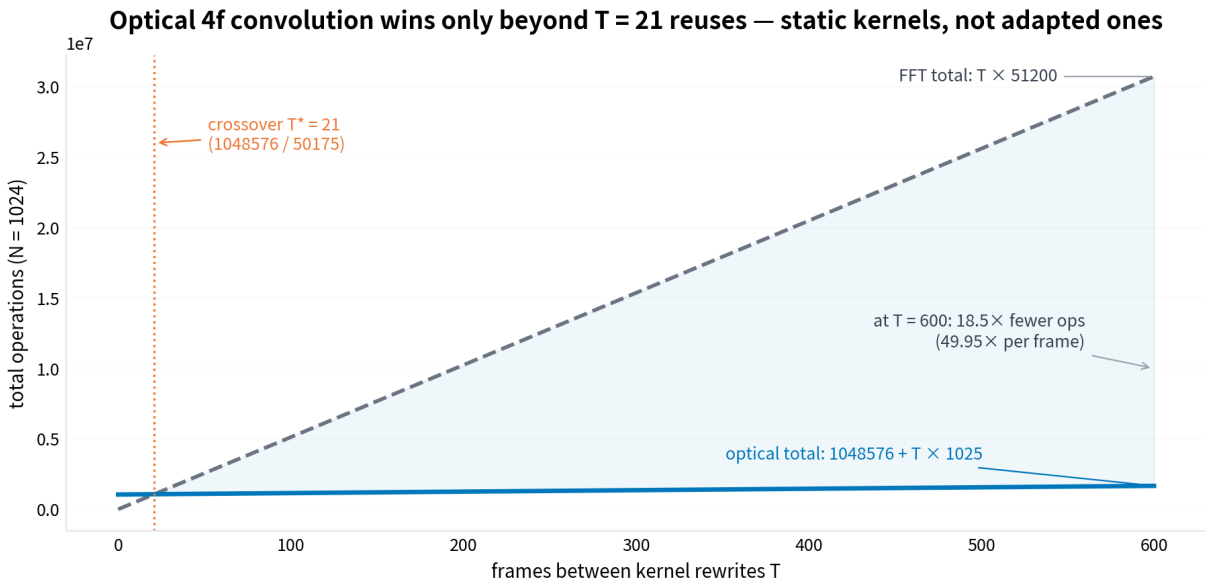
4. Decision B — a real-time convolution pipeline (fixed kernel, 60 fps)

Question. A 4K-class pipeline convolving each frame with a kernel that is re-trained every 10 seconds — 600 frames between rewrites. Optical 4f correlator or digital FFT?

Computation. Optical total = 1048576 + T·1025; FFT total = T·51200. At T = 600: optical 1,663,576 ops versus FFT 30,720,000 ops — a ratio of 3840000/207947 ≈ **18.47× fewer total operations**, and 2048/41 ≈ **49.95× per frame**. The counter-case matters as much: with a filter adapted every frame (T = 1), optical costs 1,049,601 ops versus 51,200 — **20.50× more**, because the N^2 kernel-programming setup never amortizes.

Verdict. Optical, decisively — for this reuse pattern. The crossover is T* = ceil(1048576/50175) = **21** reuses (Fig. 2): a kernel rewritten more often than every 21 frames flips the verdict to digital.

Location and caveat. The gain is amortization (precondition P2), not raw compute: the per-frame 49.95× is real only while the kernel is static. Any workload with per-frame adaptation is on the wrong side of the crossover, and the instrument below says so explicitly.



Total operations versus kernel reuse T. Optical wins beyond T = 21; at T = 600 it holds an 18.5× total-op advantage. The T = 1 counter-case (adapted filters) costs 20.5× more than FFT.

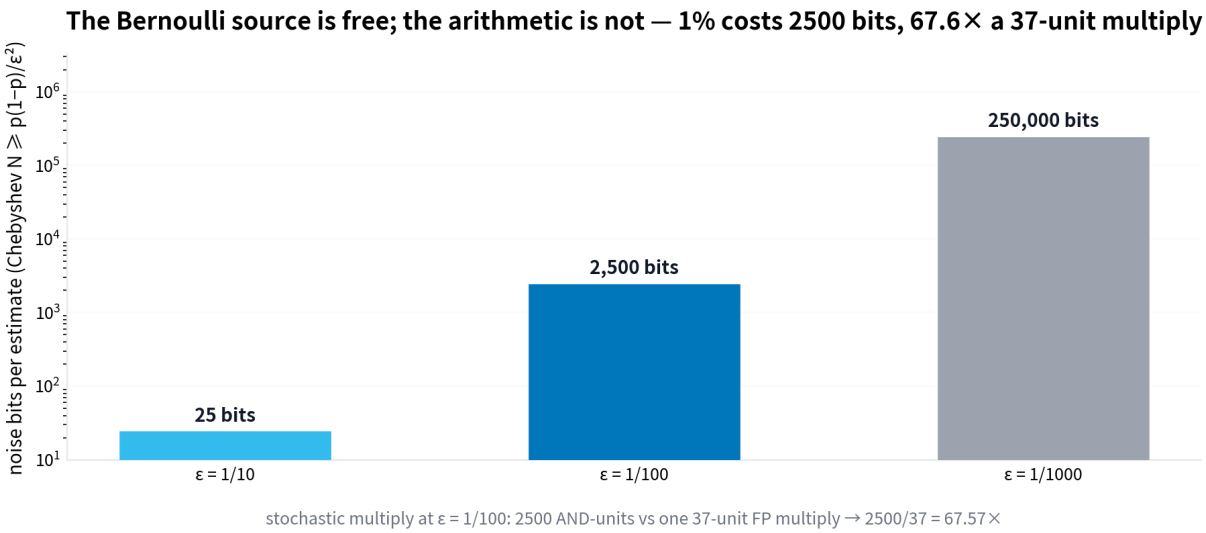
5. Decision C — Monte-Carlo estimates at 1% error

Question. Simulations need millions of Bernoulli samples at 1% error. Is a physical noise source useful here?

Computation. Chebyshev stream length $N \geq p(1-p)/\varepsilon^2$ at $p = 1/2$, by closed form and enumeration, both paths agreeing: **25 bits** at $\varepsilon = 1/10$, **2500 bits** at $\varepsilon = 1/100$, **250000 bits** at $\varepsilon = 1/1000$ (Fig. 4). The arithmetic trade: one stochastic multiply at 1% error costs 2500 AND-gate-units against one 37-unit FP multiply — **2500/37 ≈ 67.57× more energy per multiply**.

Verdict. Yes, as a randomness source; no, as an arithmetic engine. The noise stream is free where a digital PRNG costs generator state and rounding discipline — that is gain G4 of the study. But bit-serial stochastic arithmetic pays 67.6× per multiply at this tolerance, so the samples should be taken from physics and summed digitally.

Location and caveat. The crossover width (11 bits, script 03) bounds when stochastic encoding even represents the operands. And the verdict inherits the bound's pessimism: the exact binomial tail at N = 100 is far below the Chebyshev 1/4, so 2500 bits is a floor, not an estimate.



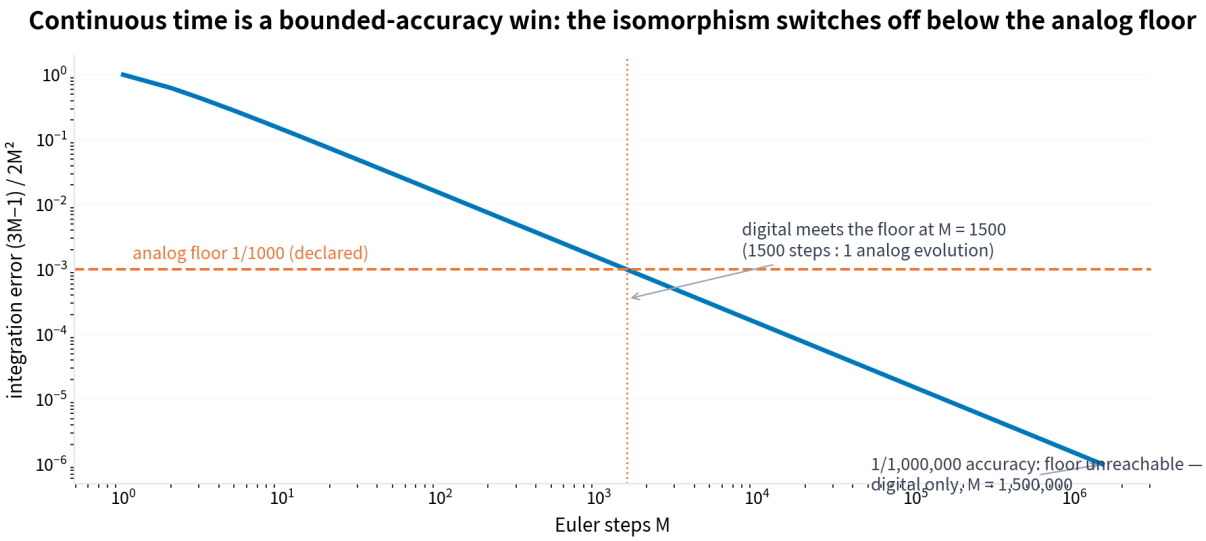
Noise bits per estimate versus error tolerance. The source is free; the arithmetic is not — 1% costs 2500 bits, 67.6× a 37-unit multiply.

6. Decision D — an analog co-processor for ODE integration

Question. A thermal-simulation loop needs y’ = f(y,t) integrated to a declared accuracy. When does one analog evolution beat digital Euler?

Computation. Euler’s error is exactly $(3M-1)/2M^2$ (verified against the direct sum for M = 1..20 in the study, re-run here). Digital reaches the analog floor 1/1000 at **M = 1500** steps — the study’s 1500:1 step ratio. Push the demand to 1/1,000,000 and digital needs **M = 1,500,000** steps — but the analog floor cannot follow: 1/1000 > 10^{-6}, so no number of analog evolutions reaches the target accuracy (Fig. 3).

Verdict. Match the floor. At $1/10^3$-class accuracy the analog co-processor is the full 1500:1 win the study measured; at $1/10^6$-class it is inadmissible, regardless of speed. Continuous time (gain G2) is a **bounded-accuracy** win, and the bound is a declared model axiom, not a measured constant — a deployment would measure its own floor first.



Euler error versus step count M against the analog floor. Digital meets the floor at M = 1500; accuracy 1/1,000,000 is unreachable in the analog model and forces digital.

7. Decision E — “should we buy an Ising annealer for our SAT workload?”

Question. The sales claim: map SAT to Ising energy and let physics find the satisfying assignment asymptotically faster.

Computation. None — and that is the answer. The study’s mapping is exact (`EQUIVALENCE = 1` on all 8 assignments of its witness instance), and its overhead is measured (16 gadget slots, 6 surviving cubic terms — the tax quadratic-only hardware pays). But exactness of encoding is not speedup: the 2^n search space maps onto 2^n energy evaluations, and the extended Church–Turing thesis stands in the study’s ledger as an **axiom**, with the quantum case explicitly open and unclaimed.

Verdict. `WOULD_BE_GUESS` — refused, with the refusal inherited directly from the study’s guardrail G-table: “Ising machines solve NP-hard problems” is rephrased to heuristics with parallel search; no class claim. What the isomorphism does license is measurable: massively parallel energy evaluation (a constant-factor story) and an exact encoding whose overhead you can budget. A purchase decision can rest on those; a complexity claim cannot.

8. What using the output bought

Decision	Verdict	Exact location	Inherited caveat
A — matmul engine	crossbar, 3.7× @ 32-bit	N ≥ 228 (32-bit) / 55 (8-bit)	converter + model axioms
B — convolution pipeline	optical, 18.5× total / 50× per frame	T* = 21 reuses	static kernels only
C — Monte-Carlo samples	noise as source, not arithmetic	2500 bits @ 1%; 67.6× per multiply	Chebyshev pessimism
D — ODE co-processor	analog at $1/10^3$, digital at $1/10^6$	M = 1500; floor 1/1000	floor is an axiom
E — Ising annealer	refused (`WOULD_BE_GUESS`)	ECT axiom; 6 cubic terms	quantum open

Three things emerged that the headline numbers alone do not give. First, **the margins are precision- and reuse-conditional**: 15× collapses to 3.7× between 8 and 32 bits; 50× per frame inverts to 20.5× worse when the kernel adapts per frame. Second, **two of the five verdicts are refusals or switch-offs** — the same instrument that says “yes, 18.5×” also says “no, not at that accuracy” and “no, that would be a guess”, exactly the guardrail behaviour the study’s table demanded. Third, **every number is reproducible**: the premises re-verified live before use, the decisions re-derivable from one script, the instrument parity-tested (§9).

9. The instrument — a substrate selector you can operate

The verdicts above are packaged as `tct_substrate_selector.html` — a single self-contained static page (inline CSS/JS, no build, no network) that runs the same five decisions interactively. Its discipline is inherited from the source study: **all comparisons are computed in integers** (cross-multiplied rational comparisons in JS), floats appear only in clearly-labelled display strings, and the Ising panel answers a “we expect exponential speedup” selection with the ECT-guardrail refusal rather than a number. The page also carries the premise table with each constant’s empirical/illustrative status, so a user cannot quote a verdict without its caveat attached.

Parity test. The page’s decision functions were extracted and executed under Node on 45 parameter sets spanning all five panels; every verdict, ratio and crossover matches `scripts/tct_applied_decisions.py` exactly (log: `tct_applied/parity.log`). The instrument is the Python engine, not a re-implementation that happens to agree.

10. Conclusions

The test succeeds, with a precise sense of “use”. The case study’s outputs — cost models, crossover formulas, guardrails, and their axiom/empirical split — function as a decision datasheet: five workload questions went in, five verdicts came out, each with its exact crossover location and inherited caveat, and nothing was claimed beyond the premises. The one general lesson: a TCT output is operational exactly where it is honest — the break-evens and the refusals are the same numbers, and an instrument built from them inherits both.